Automated Tool Wear Detection and Replacement System

MAE 564 Final Project

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1 Motivation

Tool wear is a critical factor affecting quality and productivity in CNC operations. Manufacturers traditionally rely on scheduled tool changes or manual inspection, both of which are inefficient: fixed-interval replacement wastes tool life, and while manual checks remain common in many industries, they suffer from limitations like fatigue, slow turnover, and increased labor costs.
These challenges are what motivate this tool wear detection system. With it, we hope to implement an automated inspection that can perform consistent checks and replace the tool before it affects products on the assembly line.

2 Introduction

To address the need of automated tool replacement, our project integrates **computer vision based defect detection** with a **PLC controlled replacement system**.
When a damaged or worn tool is identified, the software sends a signal to the PLC, which coordinates the removal and replacement of the part from the CNC machine. This automated workflow reduces scrap, prevents downstream equipment damage, and ensures uniform product quality.

Additionally, every step of the procedure from defect identification to tool replacement, is automated, so the need for a manual inspection is enitrely eliminated, saving labor costs.
The only human element would be replacing the spare machining tools when they run low, which can also be automated in future iterations of the system.
We did not use physical hardware for the project, rather we hoped to provide a concept of how such a system would function via MATLAB simulations and ladder-logic diagrams.

3 Methodology

We start with a computer-vision based system for estimating cutting-tool wear. Before and after each machining cycle, an image of the tool tip is analyzed to estimate wear using edge detection or erosion metrics via pixel differences. The images before and after are compared and when the wear exceeds a certain threshold, the vision module outputs a digital "Tool Worn" signal to the PLC. As we did not have access to actual CNC tools, we improvised by using images of worn and new tools to test our MATLAB code, appended.

The PLC responds by executing a tool-change sequence similar to those used in CNC automatic tool changers (ATCs): pausing the machining cycle, extracting the tool, sending the old tool to a "used tool" palette, inserting a tool from a "new tool" palette, and resuming operation.

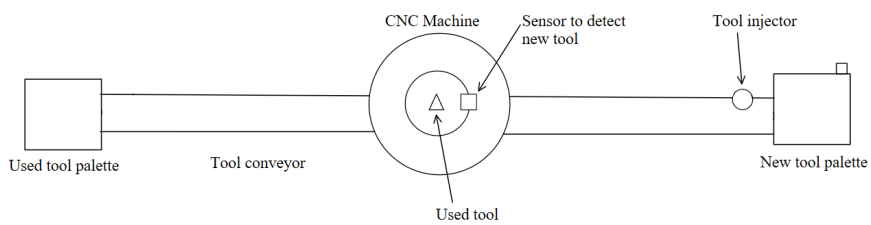


Figure 1: Schematic of PLC Controlled System

An operator will be notified of low tool supply via an alarm light that is triggered when the new tool pallet detects no tools remaining in its holsters. This will also serve to indicate the used tool palette is filled and the old tools must be disposed of before the next replacement cycle.
All aspects of the replacement cycle including the lights are controlled by a PLC, ladder logic appended.

4 Results

The project showed integration between a computer vision system and a PLC controlled sequence to tackle a known problem in the industry. The MATLAB simulation generated a binary "tool worn" signal that the PLC logic used to trigger the automated replacement routine.
This system has the potential to significantly reduce labor costs from manual checks, reduce machining errors caused by wear and improve overall machining efficiency.

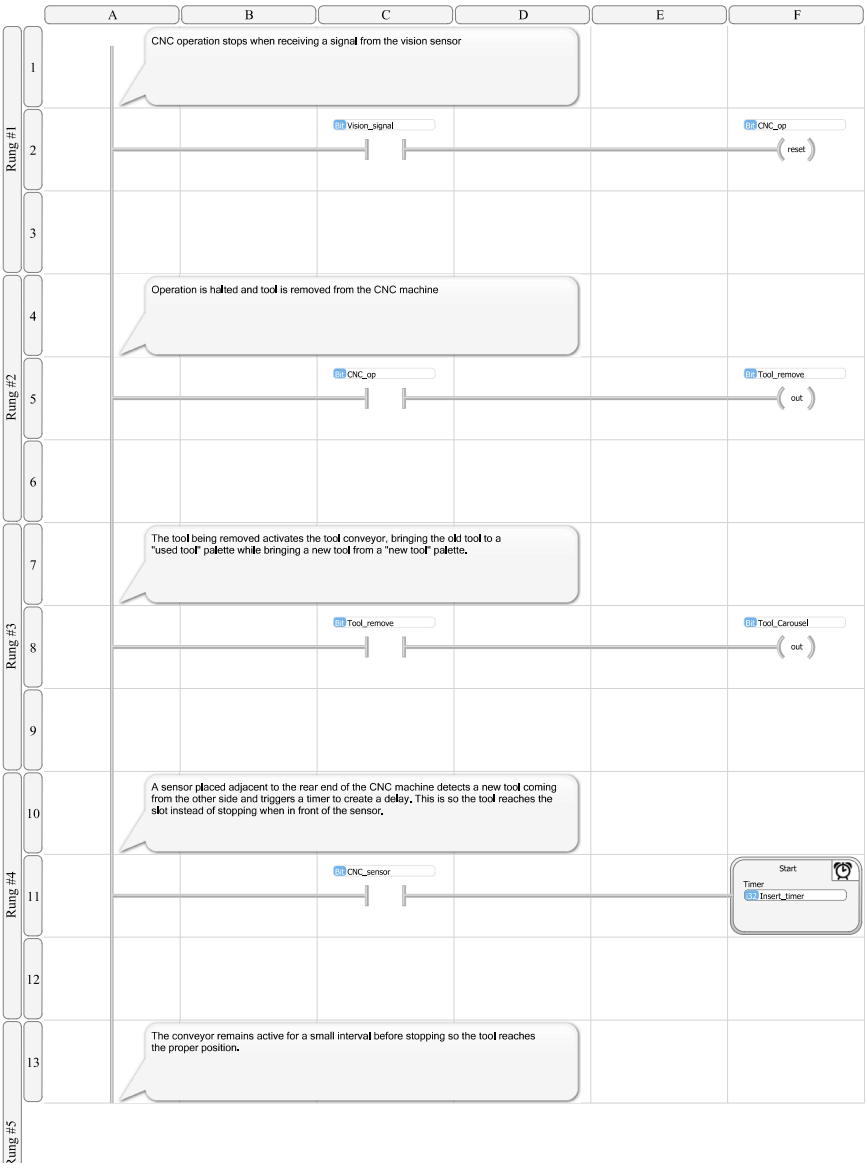
5 Appendix

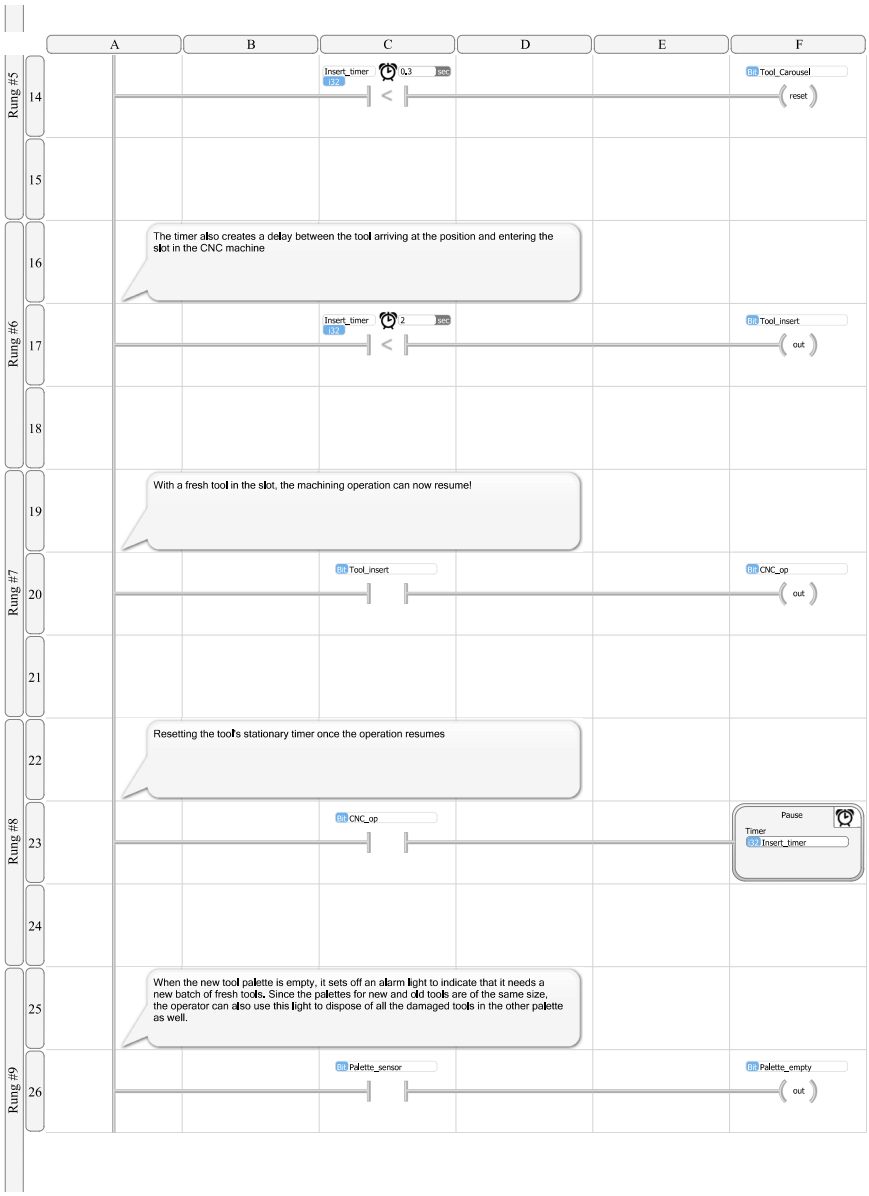
5.1 Vision Sensor Code

```

clear; clc;
% --- 1. Load Images ---
img_new  = imread('tool_new.jpg');
img_worn = imread('tool_worn.jpg');

%img_worn = imresize(img_worn, size(img_new(:,:,1)));

g_new  = rgb2gray(img_new);
g_worn = rgb2gray(img_worn);

% --- 2. Extract Tool Edge ---
E_new  = edge(g_new,  'canny',.02,8);
E_worn = edge(g_worn, 'canny',.02,8);

% --- 3. Compute Wear ---
orig_pixels    = sum(E_new(:));
current_pixels = sum(E_worn(:));
missing_pixels = orig_pixels - current_pixels;

wear_pct = missing_pixels / orig_pixels;

fprintf( Tool Wear Percentage = %.2f%%\n , wear_pct
    *100);

% --- 4. Threshold ---
TH = 0.04;
PLC_OUT = wear_pct > TH;

if PLC_OUT
    disp(  TOOL CHANGE REQUIRED );
else
    disp(  Tool OK );
end

% --- 5. Send to OpenPLC (write to MW0) ---
try
    plc = modbus('tcpip', '127.0.0.1', 502);

    % OpenPLC holds MW0 in holding register 0
    write(plc,  holdingregs , 0, uint16(PLC_OUT));

```

```
    disp( Signal sent to OpenPLC (MW0).);

catch ME
    disp( Could not connect to OpenPLC Modbus server.);
    disp(ME.message);
end

% --- 6. Visualization ---
figure;
subplot(1,2,1); imshow(E_new);  title( New Tool Edge Map );
subplot(1,2,2); imshow(E_worn); title( Worn Tool Edge Map );
    }
```

5.2 PLC Ladder Logic




A
B
C
D
E
F
Rung #5
14
Insert_timer
T2
0,3
sec
<
Tool_Carousel
reset
15
Rung #6
16
The timer also creates a delay between the tool arriving at the position and entering the slot in the CNC machine
17
Insert_timer
T2
2
sec
<
Tool_insert
out
18
Rung #7
19
With a fresh tool in the slot, the machining operation can now resume!
20
Tool_insert
CNC_op
out
21
Rung #8
22
Resetting the tool's stationary timer once the operation resumes
23
CNC_op
Pause
Timer
Insert_timer
24
Rung #9
25
When the new tool palette is empty, it sets off an alarm light to indicate that it needs a new batch of fresh tools. Since the palettes for new and old tools are of the same size, the operator can also use this light to dispose of all the damaged tools in the other palette as well.
26
Palette_sensor
Palette_empty
out